OTC-03 | ROTATING EQUIPMENT

Compressor Operation, Troubleshooting & Reliability

SUGGESTED DURATION: 3 DAYS | IN-COMPANY / LIVE ONLINE

Build a practical understanding of compressor behaviour and the evidence required to investigate trips, capacity loss, high temperatures, vibration and seal-system problems.

WHO SHOULD ATTEND

Compressor operations and maintenance teams, machinery engineers, reliability engineers and technical supervisors.

WHAT PARTICIPANTS WILL BE ABLE TO DO

- Distinguish centrifugal surge from reciprocating-compressor operating limitations.
- Interpret basic performance and condition trends before recommending action.
- Prepare a coordinated investigation involving operations, maintenance and instrumentation.

SUGGESTED PROGRAMME OUTLINE

Day 1 | Compression and operating behaviour

Gas conditions, pressure ratio, capacity and power; centrifugal performance maps and surge/choke concepts; reciprocating cylinders, valves, clearance and unloading; rod-load and reversal awareness.

Day 2 | Package systems and protection

Lube oil and cooling systems; dry gas seal and packing-system principles; separation and liquid carryover; vibration and temperature monitoring; trips, interlocks and OEM operating restrictions.

Day 3 | Troubleshooting and reliability

High discharge temperature, valve leakage, fouling and vibration; trend comparison and instrument validation; controlled shutdown decision-making; inspection findings, corrective actions and restart readiness.

OTC-03 | PRACTICAL APPLICATION & DELIVERY

Compressor Operation, Troubleshooting & Reliability

PRACTICAL WORKSHOPS

Rising vibration and seal temperature

Review an illustrative gas-compressor trend pack, agree verification steps and develop an escalation plan with defined responsibilities.

Capacity loss diagnosis

Compare centrifugal fouling symptoms with reciprocating valve leakage and distinguish the evidence needed for each machine type.

PARTICIPANT TOOLS AND TAKEAWAYS

Compressor data checklist; symptom/evidence matrix; trip-review worksheet; controlled-restart readiness checklist.

HOW LEARNING IS DELIVERED AND ASSESSED

Facilitated explanations, worked examples, case discussions and small-group exercises. A short baseline check establishes starting knowledge; a closing knowledge check and practical review assess learning. Assessment can be adapted for operators, technicians or experienced engineers.

CUSTOMISATION AND CLIENT INPUTS

Compressor type, gas composition, OEM manuals, performance data, P&IDs and selected trip trends. Live anti-surge tuning and OEM set-point changes are outside the classroom scope.

The agenda is indicative. Final content, duration, batch size, language and delivery arrangements are agreed after a training-needs discussion. Online delivery uses document reviews, calculations, role-play and case exercises; physical demonstrations require separate site arrangements.

FACILITATOR

Prem Prakash Mishra | Mechanical engineer, advisor and corporate trainer. Brings 39 years of career experience across Kuwait Oil Company, IPCL and NTPC, spanning projects, commissioning, maintenance, reliability, asset integrity and industrial audits. Specialist inputs, where needed, are agreed in the proposal.

COMPLETION AND FOLLOW-UP

An Omtech Corporation participation or completion certificate may be provided under the agreed assessment arrangements. This is company training, not an accredited professional qualification or a guarantee of certification. Post-training support is available with scope and duration agreed in the proposal.

REQUEST A CUSTOMISED TRAINING PROPOSAL

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